UNIVERSITY OF NAPLES FEDERICO II

POLYTECHNIC AND BASIC SCIENCES SCHOOL

DEPARTMENT OF INDUSTRIAL ENGINEERING

MASTER'S DEGREE IN
AUTONOMOUS VEHICLE ENGINEERING
MACHINE LEARNING AND BIG DATA

PROJECT REPORT

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1 Chosen Modelling Strategy & Motivation

The project stores the *IMDb Top-1000* film dataset in **MongoDB**, a document-oriented NoSQL database.
MongoDB was preferred because:

- **Flexible documents** – film records contain arrays (e.g. *genre*, *cast*) and optional fields (*meta_score*, *gross*). A rigid relational schema would require many join tables, whereas BSON documents can grow organically.
- **Schema-on-read** – missing values do not block loading; the Python ETL script inserts None for absent numbers and safely ignores gaps.
- **Rich aggregation pipeline** – complex calculations (group-by, unwind, sort, limit) run server-side, reducing data shipped to the client.
- **Developer productivity** – documents are JSON-like, matching the pandas→Python→MongoDB workflow in the code.

A single-collection design keeps each movie in one document, so typical analytical queries finish with one pipeline instead of multiple joins.

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2 Database Structure (Document Schema)

A *movie* document looks like:

```
{
  "title": "Inception",
  "year": 2010,
  "certificate": "PG-13",
  "runtime_minutes": 148,
  "genre": ["Action", "Adventure", "Sci-Fi"],
  "rating": 8.8,
  "meta_score": 74,        // optional
  "director": "Christopher Nolan",
  "cast": [
    "Leonardo DiCaprio",
    "Joseph Gordon-Levitt",
    "Elliot Page",
    "Tom Hardy"
  ],
  "votes": 2000000,
  "gross": 829895144       // optional
}
```

Arrays support many-to-many relationships (e.g., genres, cast).
Optional numeric fields remain absent when the CSV cell is blank; the loader converts them to integers/floats where possible.

Indexes created

Field	Reason
genre (multikey)	speeds aggregation by genre
director	used in “top directors” query
cast (multikey)	used in actor appearances
compound rating + gross	accelerates correlation analysis

The left half of **Figure 1** summarises the schema visually.

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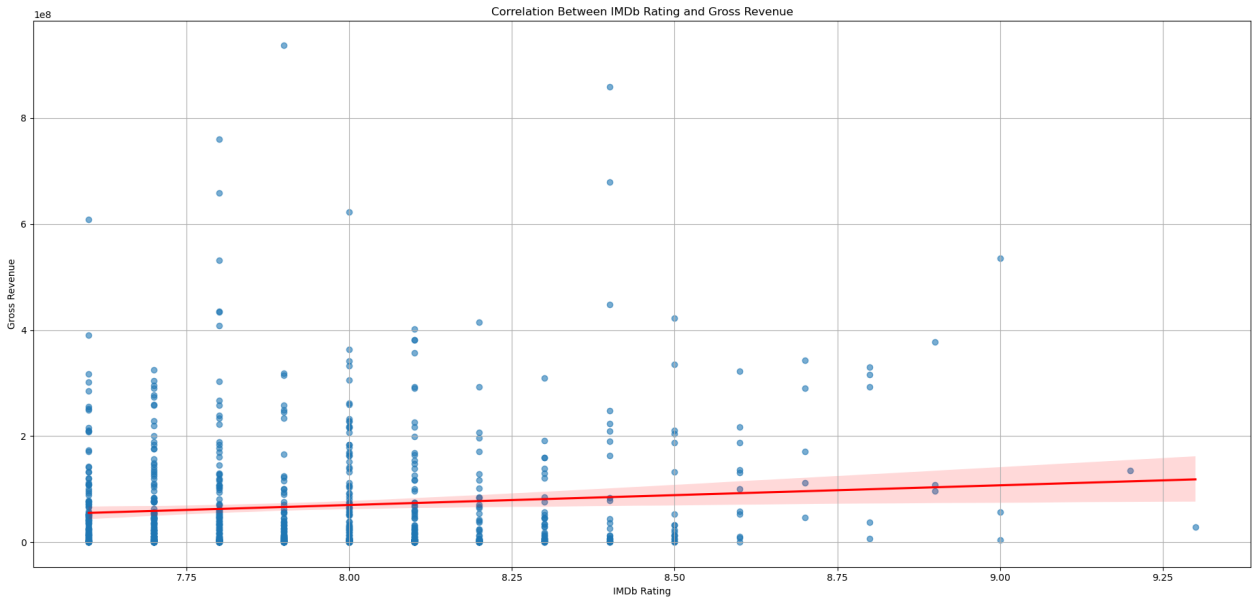
3 Implemented Queries & Key Insights

#	Aggregation pipeline	Main finding
1	unwind genre → group → count	**Drama** dominates the list (> 700 titles), followed by Comedy and Crime.
2	unwind genre → group → avg(rating)	*War*, *Western* and *Film-Noir* show the highest average ratings (~ 8.0).
3	group by director → count	**Alfred Hitchcock** (14), **Steven Spielberg** (13) and **Hayao Miyazaki** (11) are the most frequent directors in the top 1000.
4	unwind cast → group → count	**Robert De Niro** appears 17 times, ahead of Tom Hanks and Al Pacino.
5	project rating, gross → corr()	Pearson correlation between *rating* and *gross* is low-positive (~ 0.18), so critical acclaim is not a strong guarantee of box-office success.



4 Challenges & Mitigation

Challenge	Solution
Missing or malformed data (e.g. empty *gross*, commas inside numbers)	ETL step parses numbers with replace() and assigns None when parsing fails.
Comma-separated lists in CSV (genres, cast)	Split strings into Python lists before insertion.
Long labels crowd plots	Matplotlib xticks(rotation=45) plus tight_layout() for readability.
Uneven attribute presence complicates indexes	Created indexes only on commonly populated fields; sparse index option if needed.
Securing credentials in code	In production, move URI to environment variables or secrets manager; the classroom script keeps them inline for simplicity.

Conclusion

Using a document-oriented approach with MongoDB let us load, query and visualise the IMDb Top-1000 dataset rapidly. The flexible schema captured rich movie metadata; aggregation pipelines produced actionable insights, and Python notebooks rendered clear graphics. The same pattern can scale to recommendation engines or streaming-media dashboards with minimal changes.

Project Link:
https://drive.google.com/file/d/14AVbdk189z85_XyrpfiGrgd37VPTWW35/view